

Employee Attrition Prediction

Summary

A. Top 3 factors with the strongest prediction of attrition are-

1. Job role; specifically, a **Laboratory Technician**
2. Employees who do **overtime** hours
3. Employees who frequently **travel for business** purposes

B. Which department or job role should HR prioritize for retention efforts?

HR should prioritize the **Sales Department** and, specifically, the **Sales Representative** job role. The data shows that being a "Sales Representative" in the Sales Department are both strong factors of attrition.

C. Does salary alone explain attrition or are there other stronger factors?

No, salary alone does not explain attrition. While there is a correlation between low income and attrition, **work environment and role-specific demands** are stronger predictors. Factors like overtime and specific job roles (like Lab Tech) have higher coefficients than monthly income.

D. Two concrete HR recommendations

- **Stay Interviews for high-risk roles:** HR should immediately check up with all current **Laboratory Technicians** and **Sales Representatives**. Since these roles are the most likely to attrition, proactive engagement is necessary to identify their specific pain points.
- **Mandatory Overtime Review:** Since overtime is a top predictor of leaving, HR should implement a policy that requires departments to provide a business justification for any employee working overtime for more than two consecutive weeks. This forces managers to manage workloads more sustainably, reducing the risk of burnout-driven resignations.

E. What limitation does this model have?

The primary limitation is **Correlation vs. Causation**. Even though the model identifies patterns, it does not explain the reason. It is possible that Lab Techs leave because of the work environment, but it is also possible they leave because of a specific manager, a lack of equipment, or better offers from competitors that the data does not capture.

HR should treat these predictions as a signal to investigate, not as a final list. They should use the model to know where to start, but use conversations to understand the true root cause.